

Pritam Dey

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Bio

I am a statistician working on developing statistical models motivated by real-world scientific problems. My current work spans methodological advances in variational inference and symbolic regression, as well as application-driven research in spatial transcriptomics, multi-omic data integration, neuroscience, and materials science. I aim to design efficient and robust models that enable and accelerate scientific discovery.

Work Experience

Postdoctoral Research Associate, *Texas A&M University* Sep 2023–Present

Working in several collaborative teams from the Departments of Statistics and Nutrition to develop Bayesian frameworks and computational tools for spatial transcriptomics, dietary microbiome studies, and multi-omic biomarker discovery.

Graduate Research Assistant, *Duke University* Jan 2019–Aug 2024

Research Assistant developing statistical models for brain structural connectomics, including outlier detection and hierarchical continuous network representation.

Graduate Teaching Assistant, *Duke University* Aug 2019–Aug 2024

Led labs and performed grading for graduate-level courses: Linear Models (Fall 2019), Probability (Summer 2020), Probability and Measure Theory (Fall 2020), Predictive Modelling and Statistical Learning (Fall 2021), Probabilistic Machine Learning (Spring 2023).

Education

Ph.D. in Statistical Science, *Duke University* 2018–2023

- Advised by Dr. David B. Dunson.
- **Dissertation Title:** Some Advances in Statistical modeling of Brain Structural Connectomes.

M.S. in Statistical Science, *Duke University* 2018–2023

Master of Statistics, *Indian Statistical Institute, Kolkata* 2016–2018

Bachelor of Mathematics (Hons.), *Indian Statistical Institute, Bangalore* 2013–2016

- **Award:** S.H. Aravind Gold Medal for academic performance

Technical Skills

- **Statistics:** Bayesian Statistics, Variational Inference, MCMC, High-dimensional data analysis, Tree-structured modeling, Stochastic processes.
- **Bioinformatics:** Spatial transcriptomics modeling, multi-omic integration, QIIME2, PICRUSt2, edgeR.
- **Programming:** Python (Pandas, NumPy, Scikit-learn), R (tidyverse), SLURM, MATLAB, C/C++, SQL.

Publications and Preprints

Published and Accepted papers

3. **VaSST: Variational Inference for Symbolic Regression using Soft Symbolic Trees.** Roy, S., Dey, P., Mallick, B.K. *arXiv preprint. Accepted at Uncertainty in Artificial Intelligence (UAI) 2026.*
2. **Outlier detection for multi-network data.** Dey, P., Zhang, Z., Dunson, D.B. *Bioinformatics* (2022).
1. **dame-flame: a Python package providing fast interpretable matching for causal inference.** Gupta, N.R., Orlandi, V., Chang, C.-R., Wang, T., Morucci, M., Dey, P., Howell, T.J., Sun, X., Ghosal, A., Roy, S., Rudin, C., Volfovsky, A. *Journal of Statistical Software* (2025).

Preprints and papers under review

4. **A Generalized Tangent Approximation Framework for Strongly Super-Gaussian Likelihoods.** Roy, S., Dey, P., Pati, D., Mallick, B.K. *arXiv preprint. Major revision (JASA, Theory and Methods)*
3. **JASPER: Joint Bayesian Analysis of Spatial Expression via Regression.** Dey, P., Guhaniyogi, R., Ni, Y., Mallick, B.K. *arXiv preprint. Submitted*
2. **Additive Nonparametric Regression with Spatial and Network Objects.** Guhaniyogi, R., Dey, P., Chandra, K., Scheffler, A., Mallick, B.K. *Major revision (Biometrics)*
1. **Probabilistic Symbolic Regression for Equation Discovery via Operator-induced and Regularized Symbolic Forests.** Roy, S., Dey, P., Pati, D., Mallick, B.K. *arXiv preprint. Submitted*

Manuscripts in preparation

1. **Joint Integrative Spatial Transcriptomics via Bayesian Modeling for Domain Recovery and Spatially Variable Gene Selection.** Dey, P., Guhaniyogi, R., Ni, Y., Mallick, B.K. *In preparation*
2. **Ensembles of Mondrian Processes for Continuous Modeling of Structural Connectomes.** Dey, P., Zhang, Z., Dunson, D.B. *In preparation*
3. **Hierarchical Ensembles of Mondrian Processes for Simultaneous Modeling of Common and Individual Structure of connectome Data.** Dey, P., Zhang, Z., Dunson, D.B. *In preparation*
4. **Noninvasive fecal multi-omic signatures for Apc^{S580/+}/Kras^{G12D/+} mice.** Ivanov, I., Mullens, D., Dey, P., Chung, H.C., Gaynanova, I., Ni, Y., Ufondu, A., Yang, F., Han, H., Woods, P., Goldsby, J.S., Davidson, L.A., Safe, S.H., Jayaraman, A., Chapkin, R.S. *In preparation*

5. **Diet shapes gut microbial function and indirectly influences host gene expression in infants.** Dey, P., Mullens, D., Ivanov, I., Chapkin, R., Donovan, S., et al. *In preparation*

Presentations & Conferences

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| 7. Contributed talk: JSM 2026, <i>Boston</i> | 2026 (<i>Scheduled</i>) |
| 6. Invited talk: SETCASA Annual Meeting, <i>Texas A&M University</i> | 2026 |
| 5. Invited talk: IISA Conference, <i>University of Nebraska–Lincoln</i> | 2025 |
| 4. Poster: TREC Annual Cancer Research Symposium, <i>Texas A&M University</i> | 2025 |
| 3. Contributed talk: NISS 2nd Annual Graduate Student Research Conference, <i>Online</i> | 2022 |
| 2. Contributed talk: WNAR Conference, <i>Online</i> | 2022 |
| 1. Poster: Statistical Methods in Imaging Conference, <i>Online</i> | 2021 |

Awards

3. **Rank 1** in National Eligibility Test (NET), conducted by Council of Scientific and Industrial Research (CSIR), India (2018).
2. **S.H. Aravind Gold Medal**, for outstanding performance in Bachelor of Mathematics (Hons.), ISI Bangalore (2016).
1. **KVPY Scholarship**, awarded by Department of Science and Technology, Govt. of India (2013–2018).